

TAYLOR PENTZ

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 9+ years of experience designing, building, and deploying production-grade software and AI-driven systems. Extensive hands-on expertise in Python, cloud-native architectures, and deploying machine learning and generative AI solutions using modern CI/CD practices. Proven ability to design and operationalize AI/ML models including LLM-powered applications, retrieval-augmented generation (RAG), and data-intensive services in secure, scalable environments. Strong background collaborating with cross-functional stakeholders to translate complex business and healthcare-oriented requirements into reliable, measurable technical outcomes. Known for high ownership, strong engineering fundamentals, and the ability to operate effectively in regulated, enterprise-grade environments.

SKILLS

Python	JavaScript	Generative AI	Large Language Models (LLMs)
Retrieval-Augmented Generation (RAG)	Agentic AI	PyTorch	TensorFlow
Keras	Microsoft Azure	AWS	Docker
CI/CD	GitHub Actions	Jenkins	PostgreSQL
Data Modeling	Data Preprocessing	Large-Scale Datasets	Model Testing and Validation
Performance Optimization	Secure Coding	Version Control (Git)	

EXPERIENCE

Senior Software Engineer

SumatoSoft

April 2022 - December 2025

- Designed and delivered end-to-end, production-grade platforms that combined backend services, APIs, databases, and modern user interfaces, with increasing focus on AI-driven and data-intensive features.
- Built and deployed Python-based services supporting machine learning workflows, model integration, and large-scale data processing in cloud environments.
- Collaborated closely with product managers, designers, and stakeholders to translate ambiguous requirements into reliable, secure, and scalable solutions aligned with business and compliance needs.
- Implemented CI/CD pipelines using GitHub Actions and Jenkins, ensuring repeatable deployments, automated testing, and high code quality standards.
- Optimized system performance and reliability through proactive monitoring, refactoring, and architectural improvements while mentoring junior engineers and reviewing complex pull requests.
- Contributed to cloud architecture decisions and deployment strategies, emphasizing security, observability, and long-term maintainability.

Software Engineer

Hexaview Technologies Inc.

July 2018 - March 2022

- Developed and maintained full-stack applications with a strong emphasis on backend services, data persistence, and API design in production environments.
- Implemented Python-based services and integrations supporting data processing, analytics, and early-stage

machine learning use cases.

- Worked extensively with relational databases, designing schemas and optimizing queries to support high-volume, enterprise-grade applications.
- Collaborated with cross-functional teams in fast-paced client environments, balancing delivery speed with stability, security, and code quality.
- Adopted best practices in version control, testing, and CI/CD to ensure consistent releases and reduced operational risk.
- Supported production deployments and troubleshooting, gaining hands-on experience with reliability, scalability, and performance tuning.

Software Developer

SimbirSoft Company

March 2015 - June 2018

- Built and enhanced backend services and web applications supporting complex, data-heavy business workflows.
- Contributed to application logic and API development using JavaScript and Python-based stacks in real-world production systems.
- Worked closely with senior engineers to design scalable components and improve overall system architecture.
- Participated in performance optimization efforts, addressing bottlenecks and improving reliability under increasing load.
- Gained early experience with distributed systems concepts, cloud deployments, and production monitoring practices.
- Collaborated in team-based development environments, strengthening communication and software engineering fundamentals.

Software Developer Intern

Microsoft

August 2014 - February 2015

- Supported the development of internal tools and services as part of an enterprise engineering team.
- Worked with experienced engineers to implement features, fix defects, and improve code quality in production-adjacent systems.
- Gained exposure to large-scale software development practices, including version control, code reviews, and automated testing.
- Assisted with debugging and performance analysis in complex codebases used by internal stakeholders.
- Developed a strong foundation in professional software engineering standards, collaboration, and documentation.
- Built early experience working within structured, security-conscious, and highly regulated engineering environments.

EDUCATION

Bachelor of Computer Science

The University of North Carolina at Chapel Hill

2010 - 2014